R FOR DATA ANALYSIS · GETTING READY · LESSON 4

04 Open an Excel file and inspect it

Load an Excel sheet with read_excel(), then look at it with head, str, summary and colnames.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Load an Excel file with read_excel()
- Inspect a dataset with head(), str() and summary()
- List column names with colnames()

The R code in this lesson

```
library(readxl)

# Opening an Excel file
Dataset <- read_excel("Clean_Dataset.xlsx")

# Inspecting the dataset
head(Dataset)       # first 6 rows
str(Dataset)        # structure: column types
unique(Dataset)     # unique rows
summary(Dataset)    # min, max, mean for numbers
colnames(Dataset)   # all column names
print(Dataset)      # print the table
```

What you will hear (times count from the end of the intro)

Intro Now the exciting part: real data! In this lesson, you will learn to open an Excel file in R, and then look at it from different angles, so you truly understand it before any analysis. Let's open our first dataset.

0:00 Our working directory is set. Now I write a note: opening an Excel file. The line is: Dataset, arrow, read underscore excel, and the file name in quotes: Clean underscore Dataset dot x l s x.

0:12 Run it. In the Environment pane, a new item appears: Dataset, with one hundred thirty five observations of twenty four variables. That means one hundred thirty five rows, and twenty four columns.

0:25 Now let's look inside. First, head. Head shows the first six rows in the console: province, district, sex, marital status, household size, and the survey answers.

0:45 Next, str, for structure. It lists every column, and its type: chr for text, and num for numbers. This is the best way to check that each column has the right type.

1:15 Unique shows the unique rows, and summary gives quick statistics for each column.

1:35 For number columns, summary shows the minimum, the first quartile, the median, the mean, the third quartile, and the maximum. For household size, the average is about nineteen.

2:10 Then colnames, which lists all the column names. Here we see the long survey question names, like, did you receive fifty kilograms of urea?

2:35 And finally print, which prints the table in the console. I add a note above these lines: inspecting dataset.

2:50 Head, str, summary, and colnames. Use them every time you open new data. They are your first look, and your first check. Wonderful work!